

MENTAL HEALTH MONITORING SYSTEM

Phase 2 — Graph Attention Network Training

Progress Report

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1. Project Overview

The Mental Health Monitoring System is a multi-phase NLP project designed to automatically detect depression, classify mental health severity, and generate clinically coherent reports from raw text input. The system is structured as a four-layer architecture:

- Layer 1 — Transformer backbone (MentalRoBERTa) for contextual text embeddings
- Layer 2 — Graph Attention Network (GAT) for structural reasoning over token dependency graphs
- Layer 3 — Three-head classifier: emotion recognition, mental health category, and severity regression
- Layer 4 — Large Language Model (Groq) for report generation and crisis alerting

Phase 2 covers Layer 2 exclusively. The Phase 1 MentalRoBERTa backbone (fine-tuned on Dreddit then GoEmotions) is loaded and frozen for the entire notebook. Only the GAT — a symptom-level explainability layer sitting on top of the frozen embeddings — is trained in this phase.

2. Phase 2 Objective

Train a Graph Attention Network on spaCy dependency-parse graphs whose node features are frozen Phase-1 MentalRoBERTa embeddings, so that the GAT learns which grammatical relationships between tokens (negation, symptom modifiers, subject-verb) carry the strongest clinical signal for PHQ-9 depression symptoms. The trained GAT checkpoint serves as the second-layer input to the Phase 3 three-head classifier.

2.1 Revision to the Original Plan

The original Phase 2 plan assumed the 2024 PRIMATE reannotation (PRIMATE.json) covered all 9 PHQ-9 symptoms with word-level evidence spans, and that the GAT would be trained on node-level span prediction directly. After inspecting the two available files, that assumption did not hold:

File	What it actually is
primate_dataset.json	2,003 posts, each labelled yes/no across all 9 PHQ-9 symptoms. This is PRIMATE 2022 — the base dataset.
PRIMATE.json	167 entries, but only for one symptom — anhedonia (loss of interest/pleasure) — and only 41 of those 167 carry a real character-span quote. The rest are empty.

41 word-level spans is not enough to train a node-level attention objective, so the design was revised into two separate jobs:

- **Training signal (all 2,003 posts):** graph-level, 9-way multi-label classification. The GAT pools token-level node embeddings into one graph embedding per post and predicts all 9 PHQ-9 symptoms from it — this is where the model actually learns.
- **Explainability validation (the 41 spans only):** after training, the 41 anhedonia-annotated posts are run back through the trained model to check whether attention concentrates on the words a human annotator marked as evidence. This is a qualitative check — it never touches gradients and is never used as a training signal.

3. Environment and Configuration

Platform	Kaggle Notebooks
Hardware	GPU (CUDA)
Frameworks	PyTorch, PyTorch Geometric (GATConv), HuggingFace Transformers, spaCy (en_core_web_sm)
Frozen Backbone	mentalroberta_phase1_final (Phase 1 checkpoint) — hidden size 768
Max Sequence Length	512 tokens (RoBERTa hard limit)
Trainable Parameters	1,317,385 (GAT only — backbone fully frozen)

4. Dataset — PRIMATE 2022 (Base)

4.1 Overview

Source	primate_dataset.json — 2,003 Reddit posts
Task	9-way multi-label PHQ-9 symptom classification (yes/no per symptom)
Held out for explainability	41 posts (the anhedonia-span subset) — excluded from every split
Trainable posts	1,962
Train / Val / Test split	1,569 / 196 / 197 (80% / 10% / 10%)

4.2 Symptom Prevalence (of 2,003 posts)

Symptom	Positive Count	Prevalence
Worthlessness	1,680 / 2,003	83.9%
Depressed mood	1,664 / 2,003	83.1%
Fatigue	688 / 2,003	34.3%
Anhedonia	949 / 2,003	47.4%
Psychomotor	527 / 2,003	26.3%
Appetite	194 / 2,003	9.7%
Suicidal ideation	743 / 2,003	37.1%
Concentration	195 / 2,003	9.7%
Sleep	374 / 2,003	18.7%

Prevalence is highly imbalanced — worthlessness and depressed mood dominate at ~84%, while appetite and concentration appear in fewer than 10% of posts. This imbalance is corrected for during training using a per-symptom pos_weight in the loss function (Section 6.2).

5. Graph Construction Pipeline

Each post is converted into a dependency graph before it reaches the GAT:

- The frozen Phase-1 MentalRoBERTa backbone produces one embedding per subword token.
- spaCy (en_core_web_sm) parses the same post into words and a dependency tree.
- Because RoBERTa tokenizes into subwords while spaCy tokenizes into words, each spaCy word's node feature is built by averaging the embeddings of every RoBERTa subword whose character span falls inside that word — aligned exactly via the tokenizer's offset mapping.
- spaCy dependency edges (child → head) are added in both directions, plus a self-loop on every node, so information can flow both ways during message passing and a token's own features stay in the mix after aggregation.

Graph construction is the slowest step in the notebook — every post requires one forward pass through the frozen backbone. All 1,962 trainable posts were processed successfully with zero posts skipped for being too short or empty.

6. The GAT Model

6.1 Architecture

Layer 1	GATConv, 4 attention heads, concatenated, ReLU + dropout
Layer 2	GATConv, 2 attention heads, averaged (not concatenated)
Pooling	global_mean_pool — collapses node embeddings into one graph-level vector per post
Classifier head	Single linear layer to 9 logits (one per PHQ-9 symptom); sigmoid applied in the loss and at inference
Hidden dimension	256
Dropout	0.3
Trainable parameters	1,317,385 (GAT only)

Multiple attention heads in Layer 1 allow the model to learn several distinct notions of relevance at once — for example, one head attending to negations and another to symptom-adjacent adjectives — before compressing back down in Layer 2 ahead of pooling. Both layers can optionally return their attention coefficients, which is used later for the explainability check in Section 8, without changing training behaviour.

6.2 Training Configuration

Loss Function	BCEWithLogitsLoss with per-symptom pos_weight (computed from the train split only)
Optimizer	Adam, lr = 1e-3, weight decay = 1e-4
Batch Size	8
Max Epochs	30, with early stopping (patience = 5 on validation macro-F1)
Best Model Selection	Highest macro-F1 on the validation split
Optimizer Scope	GAT parameters only — the frozen backbone never appears in the optimizer

pos_weight per symptom (train split): worthlessness 0.20, depressed mood 0.20, fatigue 1.93, anhedonia 1.15, psychomotor 2.80, appetite 9.39, suicidal ideation 1.72, concentration 9.60, sleep 4.45. Rare symptoms such as appetite and concentration receive a much higher weight so the model isn't rewarded for simply predicting "no" across the board.

7. Training Progress

Epoch	Train Loss	Train F1	Val Loss	Val F1	Best?
1	0.8658	0.4375	0.8514	0.3782	✓
2	0.8332	0.4688	0.8494	0.4222	✓
3	0.8153	0.4738	0.8419	0.5021	✓
4	0.8026	0.4858	0.8765	0.3749	
5	0.7846	0.4966	0.9549	0.4130	
6	0.7715	0.5140	0.8875	0.5340	✓ (final)
7	0.7602	0.5167	0.8456	0.5008	
8	0.7433	0.5280	0.8825	0.4822	
9	0.7391	0.5363	0.9852	0.3560	
10	0.7335	0.5348	0.9314	0.5149	
11	0.7220	0.5471	0.8926	0.5270	

Training stopped early after epoch 11 (no validation improvement for 5 consecutive epochs). The best checkpoint — epoch 6, validation macro-F1 0.5340 — was reloaded for final test-set evaluation. Training loss decreases steadily throughout, but validation loss is noticeably noisier than validation F1, which is expected given the small validation set (196 posts) and the sharp class imbalance across symptoms.

8. Test-Set Evaluation

8.1 Overall Result

Test Loss	0.8428
Test Macro-F1	0.5234

8.2 Per-Symptom Breakdown

Symptom	Precision	Recall	F1	Support
Worthlessness	0.88	0.83	0.86	167
Depressed mood	0.89	0.88	0.88	167
Fatigue	0.37	0.83	0.51	66
Anhedonia	0.50	0.93	0.65	97
Psychomotor	0.55	0.49	0.52	49
Appetite	0.14	0.47	0.21	17
Suicidal ideation	0.59	0.73	0.65	75
Concentration	0.14	0.57	0.22	21
Sleep	0.23	0.18	0.20	28
Macro avg	0.48	0.66	0.52	687

Symptom	Precision	Recall	F1	Support
Weighted avg	0.66	0.78	0.69	687

8.3 Interpretation

The macro-F1 of 0.5234 hides a wide spread between symptoms. The two highest-prevalence symptoms — worthlessness (F1 0.86) and depressed mood (F1 0.88) — are predicted reliably, likely because the model has by far the most positive examples of each to learn from. The rarest symptoms — appetite (F1 0.21) and concentration (F1 0.22) — are the weakest: precision is low (0.14 for both), meaning the model over-predicts these symptoms and generates many false positives, even though pos_weight was applied specifically to counter this. Recall for these two is still reasonably high (0.47 and 0.57), so the model is not ignoring them entirely, but it is not yet precise about when they truly apply. Fatigue and anhedonia show a similar pattern of high recall (0.83, 0.93) but comparatively low precision (0.37, 0.50). Sleep is the weakest symptom overall, with both precision and recall below 0.25 — likely the hardest signal to recover from dependency structure alone. Overall, the model handles common, strongly-worded symptoms well and still struggles with the rarer, more subtly-expressed ones, which is consistent with the class-imbalance-driven challenge flagged going into training.

9. Explainability Validation — the 41 Anhedonia Spans

This step is validation only — none of the 41 posts were used for training or model selection. For each post, the trained model's second-layer GAT attention is converted into a per-node importance score, the top 20% highest-importance tokens are selected, and their character spans are compared against the human-annotated anhedonia evidence span.

Examples evaluated	41 / 41
Mean overlap — GAT attention (top 20% tokens)	15.9%
Mean overlap — random 20% token selection (20 trials/example)	14.2%
Difference (attention – random)	+1.7 percentage points
Examples with >50% overlap	4 / 41
Examples with 0% overlap	18 / 41

The GAT's attention beats a random-token baseline, but only by a modest 1.7 percentage points — some real signal is present, but it is weak, and 18 of the 41 examples show zero overlap with the human-marked evidence span. Because the training objective is graph-level across all 9 PHQ-9 symptoms rather than node-level for anhedonia specifically, nothing in training forces attention to concentrate on anhedonia-specific language, so this result is not necessarily a defect in the model. It does mean the explainability layer's word-level claims should be treated cautiously until this is investigated further — the notebook flags a manual review of a few high- and low-overlap examples as a useful next check before relying on this downstream.

10. Key Outputs and Deliverables

Deliverable	Location	Purpose
gat_phase2_best.pt	/kaggle/working/	Best checkpoint by validation macro-F1 (epoch 6), used for test evaluation and the explainability check
gat_phase2_final.pt	/kaggle/working/	Final saved checkpoint — model weights, architecture config, test macro-F1, and explainability overlap score bundled together

The final checkpoint bundles the GAT's state dict together with its architecture configuration (hidden dimension, head counts, dropout, input dimension, symptom list) plus the test macro-F1 and explainability overlap score, so it is fully self-describing for whoever loads it next in Phase 3.

11. Phase 2 Summary

Metric	Value	Notes
Trainable posts (PRIMATE 2022)	1,962	41 anhedonia-span posts held out for explainability only
Train / Val / Test split	1,569 / 196 / 197	80% / 10% / 10%
GAT trainable parameters	1,317,385	Backbone fully frozen throughout
Best validation macro-F1	0.5340	Epoch 6 — early stopping triggered at epoch 11
Test macro-F1	0.5234	Strong on common symptoms, weak on rare ones (see Section 8.3)
Explainability overlap (attention)	15.9%	vs. 14.2% random baseline — modest positive signal

12. Next Steps — Phase 3

Phase 3 takes the Phase 2 GAT checkpoint (gat_phase2_final.pt) together with the Phase 1 backbone and trains all three classifier heads end-to-end — jointly with the GAT and, per the original architecture plan, the transformer itself — using DepressionEmo and DAIC-WOZ as supervision.

- Load gat_phase2_final.pt alongside mentalroberta_phase1_final as the joint starting point
- Unfreeze the components required for end-to-end fine-tuning of the three-head classifier (emotion, mental health category, severity regression)
- Train on DepressionEmo for the emotion and category heads, and on DAIC-WOZ for the severity regression head
- Investigate the weak per-symptom precision seen on appetite, concentration, and sleep (Section 8.3) before treating those predictions as reliable downstream
- Follow up on the modest explainability overlap (Section 9) with a manual review of high- and low-overlap examples

DAIC-WOZ data use agreement should be submitted immediately if not already done — Phase 3 severity head training requires this dataset and approval can take up to 10 days.

13. Phase Status

Phase	Status	Notes
Phase 1 — Backbone Fine-Tuning	✓ COMPLETE	Checkpoint committed to Kaggle Output
Phase 2 — GAT Training	✓ COMPLETE	gat_phase2_final.pt saved; test macro-F1 0.5234
Phase 3 — Three-Head Training	PENDING	Awaiting DAIC-WOZ data use agreement
Phase 4 — LLM Integration	PENDING	Groq API integration — awaiting Phase 3 completion

